

14 Temperature

14.1 Thermal equilibrium

Candidates should be able to:

1 understand that (thermal) energy is transferred from a region of higher temperature to a region of lower temperature

2 understand that regions of equal temperature are in thermal equilibrium

14.2 Temperature scales

Candidates should be able to:

1 understand that a physical property that varies with temperature may be used for the measurement of temperature and state examples of such properties, including the density of a liquid, volume of a gas at constant pressure, resistance of a metal, e.m.f. of a thermocouple

2 understand that the scale of thermodynamic temperature does not depend on the property of any particular substance

3 convert temperatures between kelvin and degrees Celsius and recall that $T/\text{K} = \theta/°\text{C} + 273.15$

4 understand that the lowest possible temperature is zero kelvin on the thermodynamic temperature scale and that this is known as absolute zero

14.3 Specific heat capacity and specific latent heat

Candidates should be able to:

1 define and use specific heat capacity

2 define and use specific latent heat and distinguish between specific latent heat of fusion and specific latent heat of vaporisation

15 Ideal gases

15.1 The mole

Candidates should be able to:

1 understand that amount of substance is an SI base quantity with the base unit mol

2 use molar quantities where one mole of any substance is the amount containing a number of particles of that substance equal to the Avogadro constant N_A

15.2 Equation of state

Candidates should be able to:

1 understand that a gas obeying $pV \propto T$, where T is the thermodynamic temperature, is known as an ideal gas

2 recall and use the equation of state for an ideal gas expressed as $pV = nRT$, where n = amount of substance (number of moles) and as $pV = NkT$, where N = number of molecules

3 recall that the Boltzmann constant k is given by $k = R/N_A$

15.3 Kinetic theory of gases

Candidates should be able to:

1 state the basic assumptions of the kinetic theory of gases

2 explain how molecular movement causes the pressure exerted by a gas and derive and use the relationship $pV = \frac{1}{3}Nm\langle c^2\rangle$, where $\langle c^2\rangle$ is the mean-square speed (a simple model considering one-dimensional collisions and then extending to three dimensions using $\frac{1}{3}\langle c^2\rangle = \langle c_x{}^2\rangle$ is sufficient)

3 understand that the root-mean-square speed $c_{r.m.s.}$ is given by $\sqrt{\langle c^2\rangle}$

4 compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = NkT$ to deduce that the average translational kinetic energy of a molecule is $\frac{3}{2}kT$, and recall and use this expression

16 Thermodynamics

An understanding of energy from Cambridge IGCSE/O Level Physics or equivalent is assumed.

16.1 Internal energy

Candidates should be able to:

1 understand that internal energy is determined by the state of the system and that it can be expressed as the sum of a random distribution of kinetic and potential energies associated with the molecules of a system

2 relate a rise in temperature of an object to an increase in its internal energy

16.2 The first law of thermodynamics

Candidates should be able to:

1 recall and use $W = p\Delta V$ for the work done when the volume of a gas changes at constant pressure and understand the difference between the work done by the gas and the work done on the gas

2 recall and use the first law of thermodynamics $\Delta U = q + W$ expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system

